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Assignment 6 solution

Q1. solution: not

$f(x)$. $g(x)$ and $\deg(f(x))=6$ $\deg(g(x))=5$

denote that $h(x)=f(x)-g(x)$

~~according to~~ $\exists x_i$ $(i=1,2,\cdots,7)$ $x_i \neq x_j$, $i \neq j$

~~for~~ $h(x_i)=0$

But a degree n polynomial ~~only~~ has ~~almost~~ at the most n roots.

$\deg(h(x))=6$ contradiction! $\Rightarrow$ Not

Q2. solution: (b) $d=2$

Lagrange interpolation:

$$L_1(x)=\frac{(x-2)(x-3)}{(0-2)(0-3)}=\frac{1}{6}(x^2-5x+6)$$

$$L_2(x)=\frac{(x-0)(x-3)}{(2-0)(2-3)}=-\frac{1}{2}(x^2-3x)$$

$$L_3(x)=\frac{(x-0)(x-2)}{(3-0)(3-2)}=\frac{1}{3}(x^2-2x)$$

$$g(x)=1\cdot L_1(x)+3\cdot L_2(x)+0\cdot L_3(x)$$

$$=\frac{1}{6}x^2-\frac{5}{6}x+1-\frac{3}{2}x^2+\frac{9}{2}x$$

$$=-\frac{4}{3}x^2+\frac{11}{3}x+1$$

(a) $d=1$ X because $k=\frac{3-1}{2-0}=1 \neq \frac{0-3}{3-2}=-3$

(c) $d=4$ $P_4(x)=ax^4+bx^3+cx^2+dx+e$

matric form: $\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 4 & 8 & 16 \\ 1 & 3 & 9 & 27 & 81 \end{pmatrix}\begin{pmatrix} e \\ d \\ c \\ b \\ a \end{pmatrix}=\begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$

infinite number

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Q3. $(0,-2)$ $(2,1)$ $(4,4)$

$$L_1(x) = \frac{(x-2)(x-4)}{(0-2)(0-4)} \qquad y_1 = -2$$

$$L_2(x) = \frac{(x-0)(x-4)}{(2-0)(2-4)} \qquad y_2 = 1$$

$$L_3(x) = \frac{(x-0)(x-2)}{(4-0)(4-2)} \qquad y_3 = 4$$

So $g(x) = y_1L_1(x) + y_2L_2(x) + y_3L_3(x)$

$$= -\frac{2}{8}(x^2-6x+8) - \frac{1}{4}(x^2-4x) + \frac{1}{2}(x^2-2x)$$

$$= \frac{1}{4}(2x^2-4x-x^2+4x-x^2+6x-8)$$

$$= \frac{1}{4}(6x-8)$$

$$= \frac{3}{2}x - 2$$

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Q4. $(0,0)$, $(0.5, y)$, $(1,3)$ and $(2,2)$

$P_3(x)$: coefficient of x^3 is 6

Solution: Denote $P_3(x) = ~~ax^2+bx~~ 6x^3+ax^2+bx+c$

let $(0,0) \Rightarrow c=0$

$$\left.\begin{array}{l} \text{let } (1,3) \Rightarrow 6+a+b=3 \\ \text{let } (2,2) \Rightarrow 48+4a+2b=2 \end{array}\right\} \Rightarrow \begin{cases} a=-20 \\ b=17 \end{cases}$$

$P_3(x) = 6x^3-20x^2+17x$ *

$x=0.5$, so $y=\frac{17}{4}$

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